

COM3110/4115: Text Processing

Information Retrieval: Evaluating IR systems

Rob Gaizauskas

Department of Computer Science
University of Sheffield

- Definition of the information retrieval problem
- Approaches to document indexing
 - ◊ manual approaches
 - ◊ automatic approaches
- Automated retrieval models
 - ◊ boolean model
 - ◊ ranked retrieval methods (e.g. vector space model)
- Term manipulation:
 - ◊ stemming, stopwords, term weighting
- **Evaluation**
- Advanced Topics in IR

Evaluation of IR systems – Why?

- There are **various retrieval models/algorithms/IR systems**
 - ◊ How determine which is the best?
- What is the best **component/technique** for:
 - ◊ Ranking? (cosine, dot-product, ...)
 - ◊ Term selection? (stopword removal, stemming, ...)
 - ◊ Term weighting? (binary, TF, TF.IDF, ...)
- **How far down the ranked list** will a user need to look to find some/all relevant items?

- Evaluation of effectiveness carried out in relation to the **relevance** of the documents retrieved
- Relevance is judged in a **binary** way, even if it is in fact a continuous judgement
 - ◇ Impossible when the task is to **rank thousands or millions of options**: too subjective, too difficult
- Other factors could also be evaluated:
 - ◇ User effort/ease of use
 - ◇ Response time
 - ◇ Form of results presentation

Evaluation – Relevance (Benchmarking)

- In IR research/development scenarios, one cannot afford **humans** looking at results of every system/variant of system
- Instead, performance measured/compared using a pre-created **benchmarking** corpus, a.k.a. **gold-standard dataset** or **test collection**, which provides:
 - ◊ a standard set of documents, and queries
 - ◊ a list of documents judged relevant for each query, by human subjects
 - ◊ relevance scores, usually treated as binary
- Example: TREC IR evaluation corpora (<http://trec.nist.gov/>)
 - ◊ TREC (**T**ext **R**etrieval **C**onference) has run annually since 1991

Evaluation of IR systems – Metrics

- **AIM:**

1. get as much good stuff (= relevant docs) as possible
 2. get as little junk (= irrelevant docs) as possible
- The two aspects of this aim are addressed by two separate measures — **recall** and **precision**

	Relevant	Non-relevant	Total
Retrieved	A	B	A+B
Not retrieved	C	D	C+D
Total	A+C	B+D	A+B+C+D

- **Recall:** $\frac{A}{A+C}$ = proportion of relevant documents returned
- **Precision:** $\frac{A}{A+B}$ = proportion of retrieved documents that are relevant
 - ◇ Both measures have **range**: $[0 \dots 1]$

Another Terminology

- Another terminology speaks of **true/false** + **positives/negatives**
 - ◇ **positives/negatives** $\rightsquigarrow$ retrieved/not retrieved
 - ◇ **true/false** $\rightsquigarrow$ whether item is **correctly** positive/negative
 - ◇ more general, i.e. used in other contexts, such as medical testing
 - e.g., a test returns a false positive result if it signals the presence of a disease when in fact the disease is not present

- Previous:

	Relevant	Non-relevant
Retrieved	A	B
Not retrieved	C	D

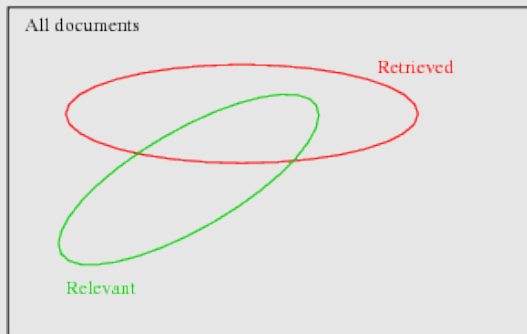
maps as:

- ◇ A : **true positives** (TP)
- ◇ B : **false positives** (FP)
- ◇ C : **false negatives** (FN)
- ◇ D : **true negatives** (TN)

- **Recall:** $\frac{TP}{TP+FN}$ **Precision:** $\frac{TP}{TP+FP}$

Retrieved vs. Relevant Documents

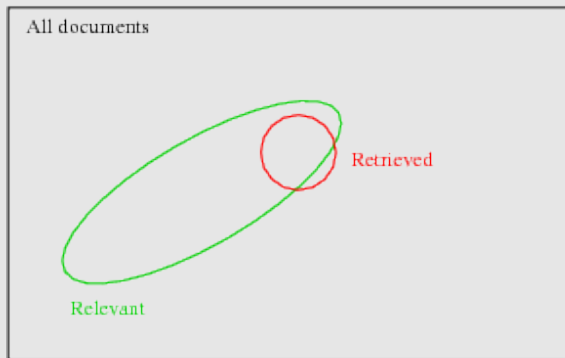
- Precision and Recall address the relation between the *retrieved* and *relevant* sets of documents



- Various situations that arise can be pictorially represented in these terms

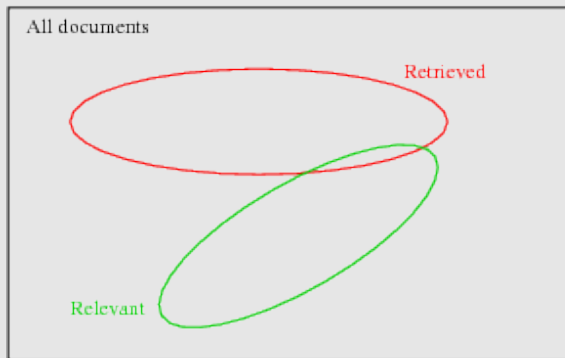
Retrieved vs. Relevant Documents (contd)

- High precision, low recall:



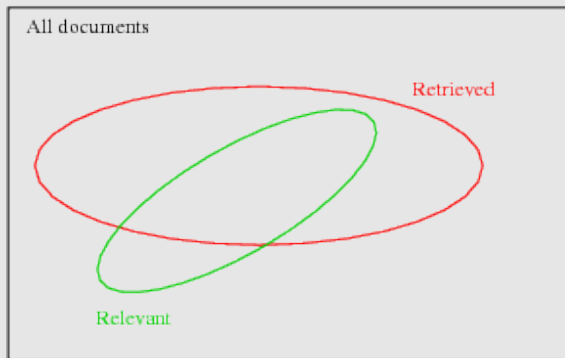
Retrieved vs. Relevant Documents (contd)

- Low precision, low recall:



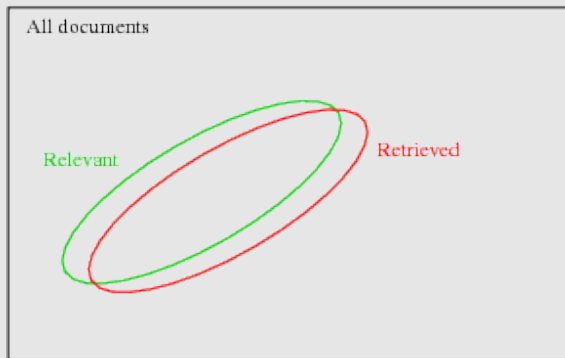
Retrieved vs. Relevant Documents (contd)

- Low precision, high recall:



Retrieved vs. Relevant Documents (contd)

- High precision, high recall:

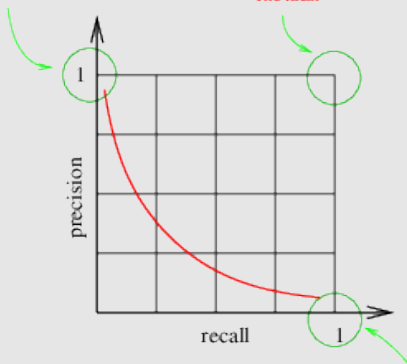


Trade-off Between Recall and Precision

- There is always a trade-off between precision and recall
 - ◇ For IR: as more results are considered down the list, precision generally drops, while recall generally increases

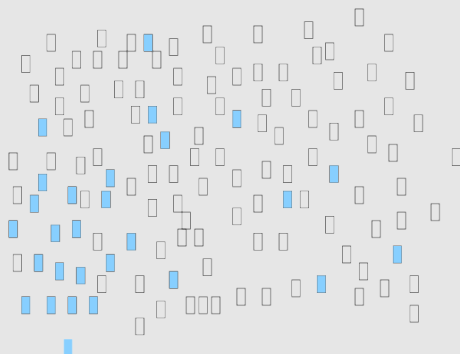
Returns relevant documents but
misses many useful ones too

The ideal



Returns most relevant documents
but includes lots of junk too

Recall and Precision: Example



	Rel	Non-rel
Ret	A	B
Not ret	C	D

- All documents: $A+B+C+D = 130$
- Relevant documents for query: $A+C = 28$ (blue boxes)

Recall and Precision: System 1



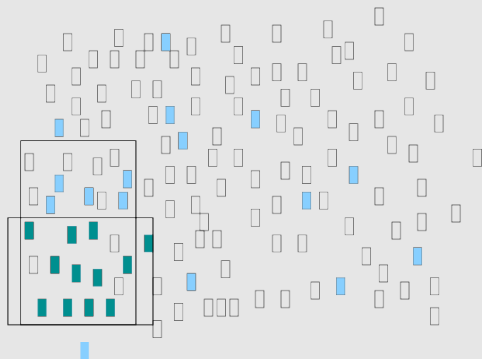
	Rel	Non-rel
Ret	A	B
Not ret	C	D

$$R_1 = \frac{A_1}{A_1 + C_1} = \frac{16}{28} = .57$$

$$P_1 = \frac{A_1}{A_1 + B_1} = \frac{16}{25} = .64$$

- System 1 retrieves 25 items: $A_1 + B_1 = 25$ (those inside rectangle)
- Relevant and retrieved items: $A_1 = 16$ (green boxes)
- Relevant documents for query: $A_1 + C_1 = 28$ (green + blue boxes)

Recall and Precision: System 2



	Rel	Non-rel
Ret	A	B
Not ret	C	D

$$R_2 = \frac{A_2}{A_2 + C_2} = \frac{12}{28} = .43$$

$$P_2 = \frac{A_2}{A_2 + B_2} = \frac{12}{15} = .8$$

- System 2 retrieves 15 items: $A_2 + B_2 = 15$ (in horizontal rectangle)
- Relevant and retrieved items: $A_2 = 12$ (green boxes)
- Relevant documents for query: $A_2 + C_2 = 28$ (green + blue boxes)

Recall and Precision: Which is the better system?



$$R_1 = \frac{A_1}{A_1 + C_1} = \frac{16}{28} = .57$$
$$P_1 = \frac{A_1}{A_1 + B_1} = \frac{16}{25} = .64$$

$$R_2 = \frac{A_2}{A_2 + C_2} = \frac{12}{28} = .43$$
$$P_2 = \frac{A_2}{A_2 + B_2} = \frac{12}{15} = .8$$

- Which did better: System 1 or System 2?

- F measure (also called $F1$):

- ◇ combines precision and recall into a single figure
- ◇ gives equal weight to both:

$$F = \frac{2PR}{P + R}$$

- F is a **harmonic mean**:

- ◇ penalises low performance in one value more than **arithmetic** mean:

	values	mean	F
e.g.	P=0.5, R=0.5	0.5	0.5
	P=0.1, R=0.9	0.5	0.18

- ◇ Previous example:

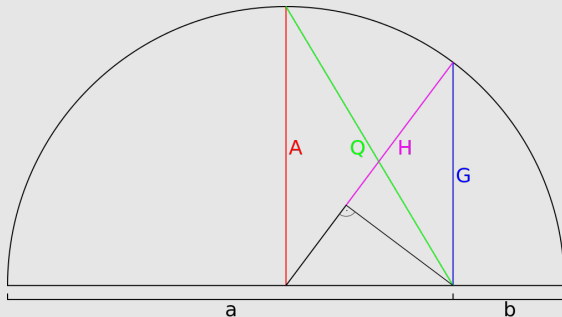
	R	P	F
System 1	.57	.64	0.603
System 2	.43	.8	0.559

- Related generalised measure F_β :

$$F_\beta = \frac{(1 + \beta^2) \cdot (P \cdot R)}{(\beta^2 \cdot P) + R}$$

- ◇ allows user to determine relative importance of P vs. R , by *varying* β
- ◇ β is chosen such that recall is considered β times as important as precision, is
- ◇ $F1$ is a *special case* of F_β , where $\beta = 1$

Aside on Pythagorean Means



(from: https://en.wikipedia.org/wiki/Harmonic_mean)

- A=Arithmetic mean of scalars 'a' and 'b': $\frac{a+b}{2}$
- G=Geometric mean: $\sqrt[2]{a \cdot b}$
- H=Harmonic mean: $\frac{2}{\frac{1}{a} + \frac{1}{b}}$
- Q=Quadratic mean or Root mean square: $\sqrt{\frac{1}{2}(a^2 + b^2)}$

Precision at a cutoff

- Measures how well a method **ranks relevant documents** before non-relevant documents
- E.g. there are **5 relevant documents = d1,d2,d3,d4,d5** – compute precision at top 5

	System 1	System 2	System 3
<i>rank 5:</i>	d1: ✓	d10: ✗	d6: ✗
	d2: ✓	d9: ✗	d1: ✓
	d3: ✓	d8: ✗	d2: ✓
	d4: ✓	d7: ✗	d10: ✗
	d5: ✓	d6: ✗	d9: ✗
<i>rank 10:</i>	d6: ✗	d1: ✓	d3: ✓
	d7: ✗	d2: ✓	d5: ✓
	d8: ✗	d3: ✓	d4: ✓
	d9: ✗	d4: ✓	d7: ✗
	d10: ✗	d5: ✓	d8: ✗

precision at rank 5:

1.0

0.0

0.4

precision at rank 10:

0.5

0.5

0.5

Precision at a cutoff (ctd)

- Note precision at top 5 for System 1: inner order of relevant documents doesn't matter as long as they are all relevant

	System 1	System 2	System 3
<i>rank 5:</i>	d5: ✓	d10: ✗	d6: ✗
	d4: ✓	d9: ✗	d1: ✓
	d3: ✓	d8: ✗	d2: ✓
	d1: ✓	d7: ✗	d10: ✗
	d2: ✓	d6: ✗	d9: ✗
<i>rank 10:</i>	d6: ✗	d1: ✓	d3: ✓
	d7: ✗	d2: ✓	d5: ✓
	d8: ✗	d3: ✓	d4: ✓
	d9: ✗	d4: ✓	d7: ✗
	d10: ✗	d5: ✓	d8: ✗
<i>precision at rank 5:</i>	1.0	0.0	0.4
<i>precision at rank 10:</i>	0.5	0.5	0.5

Average Precision

- Aggregates many precision numbers into one evaluation figure
- Precision computed for each point a relevant document is found, and figures averaged

System 1		System 2		System 3	
d1:	✓ (1/1)	d10:	×	d6:	×
d2:	✓ (2/2)	d9:	×	d1:	✓ (1/2)
d3:	✓ (3/3)	d8:	×	d2:	✓ (2/3)
d4:	✓ (4/4)	d7:	×	d10:	×
d5:	✓ (5/5)	d6:	×	d9:	×
d6:	×	d1:	✓ (1/6)	d3:	✓ (3/6)
d7:	×	d2:	✓ (2/7)	d5:	✓ (4/7)
d8:	×	d3:	✓ (3/8)	d4:	✓ (5/8)
d9:	×	d4:	✓ (4/9)	d7:	×
d10:	×	d5:	✓ (5/10)	d8:	×

<i>precision at rank 5:</i>	1.0	0.0	0.4
<i>precision at rank 10:</i>	0.5	0.5	0.5
<i>avg. prec:</i>	1.0	0.354	0.573

Summary

- **Evaluation** of IR systems necessary to compare systems or system variants with each other to determine which is best
- Typically done by measuring ability of system to return **relevant** documents using a **gold-standard test collection**, comprising:
 - ◇ a set of documents
 - ◇ a set of queries
 - ◇ a set of human binary relevance judgements indicating which documents are relevant to each query
- Standard evaluation measures employed are:
 - ◇ **precision** – proportion of documents retrieved that are relevant
 - ◇ **recall** – proportion of relevant documents that are retrieved
 - ◇ **f-measure** - harmonic mean of recall and precisions
- To assess how well a system ranks relevant documents ahead of non-relevant documents use **precision at a cut-off** or **average precision**